

MTE 203 – Advanced Calculus

Week 2 Tutorial Solutions

Problem 1 [S. 11.5, P.7]

Find the equation of the plane containing the line,

$$x - y + 2z = 4$$

$$2x + y + 3z = 6$$

and the point $(1, -2, 4)$.

Solution:

Note that the equation of line in this problem is not given directly. It is given as the intersection of two planes. Hence, a vector along the line is given by the cross product of the two normal vectors to the planes.

The normal vectors to each of the given planes are:

$$\begin{aligned}\vec{n}_1 &= (1, -1, 2) \\ \vec{n}_2 &= (2, 1, 3)\end{aligned}$$

A vector along the line is therefore given by,

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = (1, -1, 2) \times (2, 1, 3) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ 2 & 1 & 3 \end{vmatrix}$$

$$\vec{v} = (-5, 1, 3)$$

A point in the line can be obtained by equating (or intersecting) both planes:

$$x - y = 4 - 2z$$

$$2x + y = 6 - 3z$$

Then,

$$3x = 10 - 5z$$

$$x = \frac{10 - 5z}{3}$$

We can arbitrarily choose $z = 0$, and find that

$$x = \frac{10}{3}$$

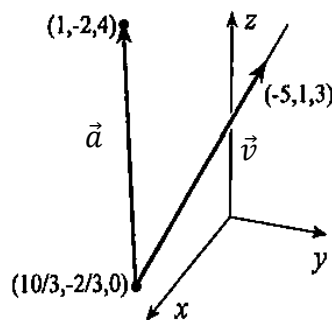
And

$$y = -\frac{2}{3}$$

Therefore, a point in the line is,

$$\left(\frac{10}{3}, -\frac{2}{3}, 0\right)$$

From the figure, since both $(1, -2, 4)$ and $\left(\frac{10}{3}, -\frac{2}{3}, 0\right)$ are points in the required plane, a second vector $\vec{a}$ in this plane can be obtained as,



$$\vec{a} = (1, -2, 4) - \left(\frac{10}{3}, -\frac{2}{3}, 0\right) = \left(-\frac{7}{3}, -\frac{4}{3}, 4\right)$$

Or multiplying all components of these vectors by 3 (to avoid working with fractions), we obtain,

$$\vec{a} = (-7, -4, 12)$$

A vector normal to the required plane will then be,

$$(-5, 1, 3) \times (-7, -4, 12) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -5 & 1 & 3 \\ -7 & -4 & 12 \end{vmatrix} = (24, 39, 27)$$

Or the equivalent parallel vector

$$\vec{n} = (8, 13, 9)$$

The equation of the plane can then be written as,

$$(8, 13, 9) \cdot (x - 1, y + 2, z - 4) = 0$$

$$8x + 13y + 9z = 18$$

Problem 2 [S. 11.6, P.27]

Find the distance between the skewed lines¹ defined by the intersection of the planes,

$$x + y - z = 4$$

$$2x - z = 4$$

And the line,

$$x = \frac{y + 1}{2} = \frac{z - 1}{3}$$

Solution:

The direction of the first line can be obtained from the cross product of the two normal vectors of the two planes as:

$$(1, 1, -1) \times (2, 0, -1) = (-1, -1, -2)$$

Or we can multiply this direction by -1,

¹ The distance between skew lines was solved in Lecture 3 and it is also solved in Example 24 of your textbook.

$$\vec{v}_1 = (1, 1, 2)$$

A point in the line of intersection can be found by equating the two planes,

$$\begin{aligned}x + y &= 4 + z \\2x &= 4 + z\end{aligned}$$

$$\begin{aligned}x - y &= 0 \\x &= y\end{aligned}$$

If $z = 0$,

$$\begin{aligned}2x &= 4 \\x &= 2\end{aligned}$$

And the point is,

$$R = (2, 2, 0)$$

The parametric form of the equation of the line of intersection of the two planes is,

$$\begin{aligned}x &= 2 + s \\y &= 2 + s \\z &= 2s\end{aligned}$$

The parametric form of the second line is

$$\begin{aligned}x &= t \\y &= -1 + 2t \\z &= 1 + 3t\end{aligned}$$

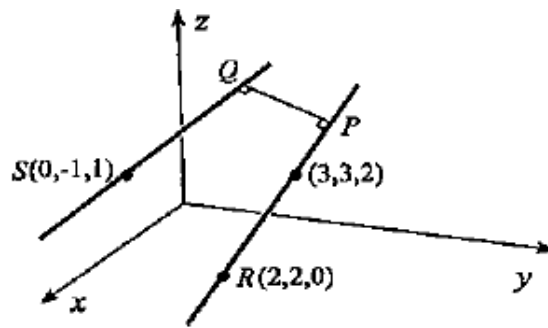
A point in the second line is

$$(0, -1, 1)$$

and the direction of the second line is,

$$\vec{v}_2 = (1, 2, 3)$$

Let us draw both skew lines,



As determined in class, **the distance between two skew lines**, is given by the component of $\overrightarrow{RS}$, which is formed between two arbitrary points selected on the two lines, along $\overrightarrow{PQ}$, which is a line segment perpendicular to both lines. Hence, this distance is given by,

$$d = |\overrightarrow{RS} \cdot \hat{n}|$$

The vector $\overrightarrow{RS}$ is,

$$\overrightarrow{RS} = S - R = (0, -1, 1) - (2, 2, 0) = (-2, -3, 1)$$

A vector along the line $\overrightarrow{PQ}$ can be found by crossing the direction vectors of both lines,

$$\vec{n} = (1, 1, 2) \times (1, 2, 3) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix} = (-1, -1, 1)$$

And

$$\hat{n} = \frac{(-1, -1, 1)}{\sqrt{3}}$$

The distance between the two lines is then,

$$d = \left| (-2, -3, 1) \cdot \frac{(-1, -1, 1)}{\sqrt{3}} \right| = \frac{6}{\sqrt{3}}$$